Analyses

H1: policy vs hull
- 27 primary tests
- 30 secondary tests
- 10 stronger policies

Robustness
- 3 grading rules
- 5 cost axes
- single plan baseline
- margin test at 0.03

H3: multi step
- IRCoT vs the hull
- 3 readers x 3 sets

Mechanism
- oracle headroom
- what planners learn

Weak protocol
- Always RAG baseline
- one seed, strict EM

Protocol

hull fitted on training folds | 5 folds x 5 seeds | t test, bootstrap CI | Holm per family | must pass in both plan sets

Policies

chosen per question

Primary
- TSR planner
- Adaptive RAG
- Adaptive k
- Closed book gate
- Confidence cascade
- Reader cascades

Secondary
- Adaptive k, rule
- AR, BGE M3
- AR, FLAN T5
- Per plan LR
- Per plan kNN
- METIS port

Stronger
- Classifier gate
- Planner with confidence

Reference
- IRCoT multi step

Measurement

Grade every answer
- label_final: strict rule, Qwen3 14B judge if ambiguous
- human audit: 2 x 200 answers, 3 independent raters

Cost of every answer
- mean measured latency per question (main axis)
- tokens, output priced 4x, FLOPs, calls

Plan matrix

every question x every plan x every reader

6 retrievers x depth 0 to 20
- BM25
- SPLADE v3
- BGE M3
- Stella 1.5B
- BM25 + BGE M3 (RRF)
- Qwen3 Reranker 0.6B

Two hop plans
- BM25, 3 or 5 per hop
- reader writes the second query

5 readers, greedy
- Qwen2.5 7B, bf16
- Llama 3.1 8B, bf16
- Qwen3 14B, int8
- Gemma 3 27B, 4 bit
- Qwen3 32B, 4 bit

Data

4 QA datasets, 500 questions each
- Natural Questions, HotpotQA (one and two hop)
- 2WikiMultihopQA, MuSiQue (multi hop)

Corpora
- BEIR Wikipedia for NQ (2.7M) and HotpotQA (5.2M)
- pooled paragraphs for 2Wiki and MuSiQue